

Implementing the Bayes Classifier for Linearly Separable Multivariate Data

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Abstract—This project delves into the implementation of a ****Bayesian classifier**** using multivariate Gaussian distributions to classify data points into one of three distinct classes. The classifier computes posterior probabilities based on Bayes' theorem and is evaluated using various statistical metrics. Additionally, decision boundaries are visualized to demonstrate the model's effectiveness in separating the feature space into distinct regions corresponding to the classes. Through rigorous evaluation, this project highlights the strengths of the Bayesian classifier in statistical pattern recognition.

We provide hands-on experience in developing, evaluating, and visualizing a Bayesian classifier, while also reinforcing theoretical concepts such as Gaussian distribution assumptions, class priors, and posterior probability calculations.

I. INTRODUCTION

The ****Bayesian classifier**** is a powerful tool in statistical pattern recognition. In this project, we use it to classify data points into one of three classes based on their features. Each class is assumed to follow a multivariate Gaussian distribution. The classifier calculates the posterior probability for each class and assigns data points to the class with the highest posterior.

This project aims to: - Implement a Bayesian classifier using multivariate Gaussian distributions. - Evaluate the classifier's performance with metrics such as accuracy, precision, recall, and F1-score. - Visualize the decision boundaries that separate the classes, providing insight into the classifier's behavior.

The classifier is tested on a dataset where each class is represented by two features. Through analysis and visualization, we assess how well the classifier distinguishes between classes.

$$P(C_k | x) = \frac{P(x | C_k) \cdot P(C_k)}{P(x)}$$

- **Posterior Probability** $P(C_k | x)$: Probability of class C_k given the observed data x .
- **Prior Probability** $P(C_k)$: Initial belief about class C_k before observing the data.
- **Likelihood** $P(x | C_k)$: Probability of observing data x given that the class is C_k .
- **Evidence** $P(x)$: Normalizing constant ensuring the sum of posterior probabilities equals 1.

II. OBJECTIVES

The key objectives of this project are:

- 1) **Classifier Development**: Build a Bayesian classifier based on multivariate Gaussian distributions for three classes of data.
- 2) **Model Evaluation**: Use statistical metrics such as confusion matrix, accuracy, precision, recall, and F1-score to evaluate the classifier's performance.
- 3) **Decision Boundary Visualization**: Visualize the decision boundaries that separate the different classes, allowing for a clear understanding of how the classifier behaves.

III. EXPLORATORY DATA ANALYSIS (EDA)

Before implementing the classifier, it is essential to explore and understand the data. We load the data, which consists of three classes, each with two features. Using scatter plots, we can observe how each class is distributed and identify any overlap or separability.

A. Data Distribution

Distribution plots are used to visualize the distribution of the data points for each class in the feature space. This helps identify any overlaps or separability between classes.

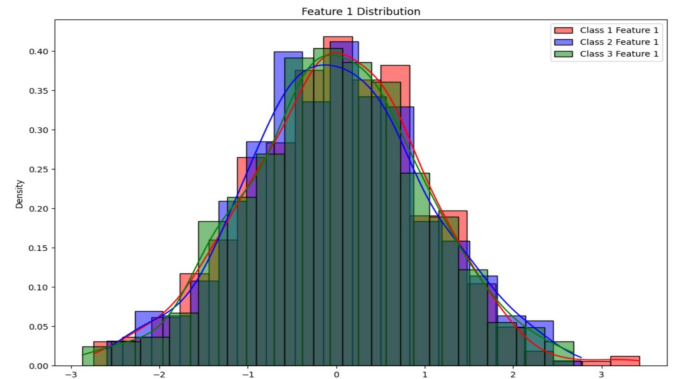


Fig. 1. Showing class distribution.

B. Cluster Formation

The distribution plots also help reveal any natural clustering of data points within each class, providing insight into the structure of the dataset.

IV. DATA PROCESSING

The data is preprocessed to ensure it is suitable for the Bayesian classifier. This involves several key steps:

A. Normalization/Standardization

The data is standardized to have zero mean and unit variance to ensure that the features are on a comparable scale.

B. Data Splitting

The dataset is split into training and testing sets to evaluate the classifier's performance. The training set is used to build the model, while the test set is used for evaluation.

V. BAYESIAN CLASSIFIER

A. Multivariate Gaussian Distribution

We compute the mean and covariance matrix for each class. The classifier assumes that the data follows a multivariate Gaussian distribution, which is characterized by the following probability density function:

$$f(x; \mu, \Sigma) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right)$$

where:

- x is the data point,
- μ is the mean vector,
- Σ is the covariance matrix.

B. Posterior Calculation

The posterior probability for each class is calculated using Bayes' theorem. The class with the highest posterior is assigned to the data point.

VI. DIAGONAL COVARIANCE MATRIX

In this scenario, the covariance matrix for each class is diagonal, meaning that the off-diagonal elements, which represent correlations between different features, are zero. The diagonal elements represent the variances of each feature. This assumption simplifies the model as it reduces the number of parameters to be estimated, making it computationally more efficient. This approach assumes that the features are uncorrelated with each other within each class.

The probability density function for a multivariate Gaussian distribution with a diagonal covariance matrix is given by:

$$f(x; \mu, \Sigma) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right)$$

where Σ is a diagonal matrix specific to each class.

VII. SHARED FULL COVARIANCE MATRIX

In this case, all classes share the same full covariance matrix, denoted as Σ . This assumption implies that the shapes of the distributions for different classes are identical, though their centers (means) might differ. The covariance matrix Σ contains non-zero off-diagonal elements, allowing the model to capture correlations between features across all classes.

The probability density function for this scenario is:

$$f(x; \mu_k, \Sigma) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu_k)^T \Sigma^{-1} (x - \mu_k) \right)$$

where the covariance matrix Σ is shared across all classes, simplifying the model by assuming similar variances and covariances for all classes.

VIII. DIAGONAL COVARIANCE MATRIX FOR EACH CLASS

Under this assumption, the covariance matrix is diagonal, but each class has its own distinct diagonal matrix. This means that while the features are uncorrelated within each class, the variances of the features differ from class to class. The diagonal elements of the covariance matrix represent the variance of each feature, and these variances can vary across classes.

The probability density function for each class is:

$$f(x; \mu_k, \Sigma_k) = \frac{1}{(2\pi)^{d/2} |\Sigma_k|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu_k)^T \Sigma_k^{-1} (x - \mu_k) \right)$$

where Σ_k is the diagonal covariance matrix specific to class k .

IX. FULL COVARIANCE MATRIX FOR EACH CLASS

In this approach, each class has its own full covariance matrix Σ_k , which allows the model to capture the specific correlations between features for each class. This is the most flexible and general form of the multivariate Gaussian distribution, as it makes no assumptions about the relationships between features being the same across classes.

The probability density function for this model is:

$$f(x; \mu_k, \Sigma_k) = \frac{1}{(2\pi)^{d/2} |\Sigma_k|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu_k)^T \Sigma_k^{-1} (x - \mu_k) \right)$$

where Σ_k represents the full covariance matrix for class k , capturing unique variances and covariances for each class.

X. DIAGONAL COVARIANCE MATRIX

In this configuration, each class is assumed to have a diagonal covariance matrix, meaning there are no correlations between features, and the variance is only captured along the diagonal of the covariance matrix. This assumption simplifies the model and is computationally efficient, but it may not capture the true nature of the data if features are correlated.

A. Results and Observations

- **Precision per class:** [0.9934, 0.9868, 1.0000]
- **Recall per class:** [1.0000, 0.9933, 0.9867]
- **Mean Recall:** 0.9933
- **F-Measure per class:** [0.9967, 0.9900, 0.9933]
- **Mean F-Measure:** 0.9933
- **Classifier Accuracy:** 99.33%

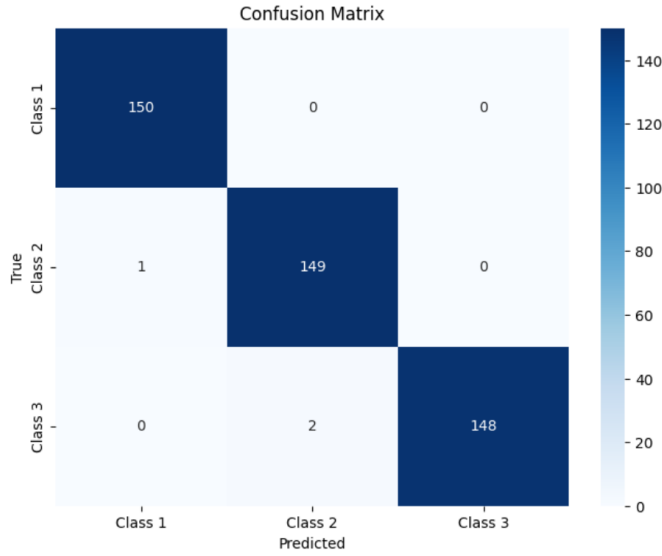


Fig. 2. Confusion Matrix (Diagonal Covariance Matrix)

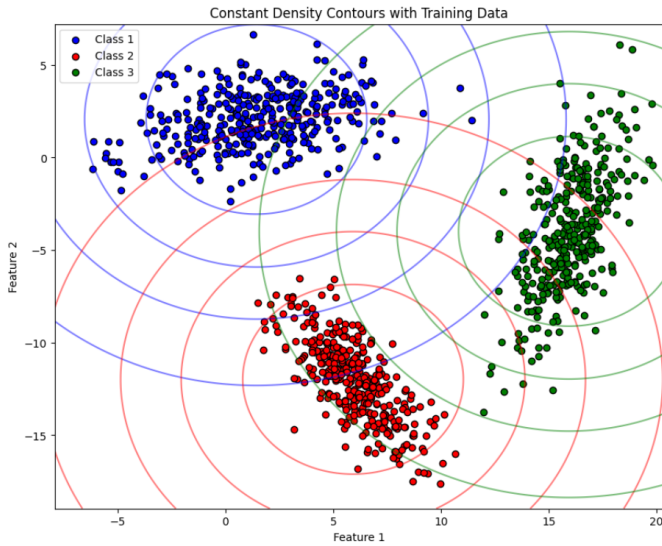


Fig. 3. Constant Density Contours with Training Data

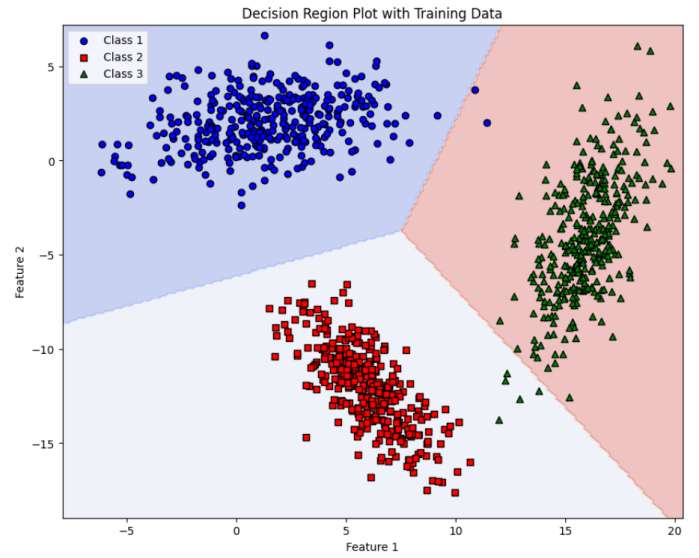


Fig. 4. Decision Region Plot with Training Data

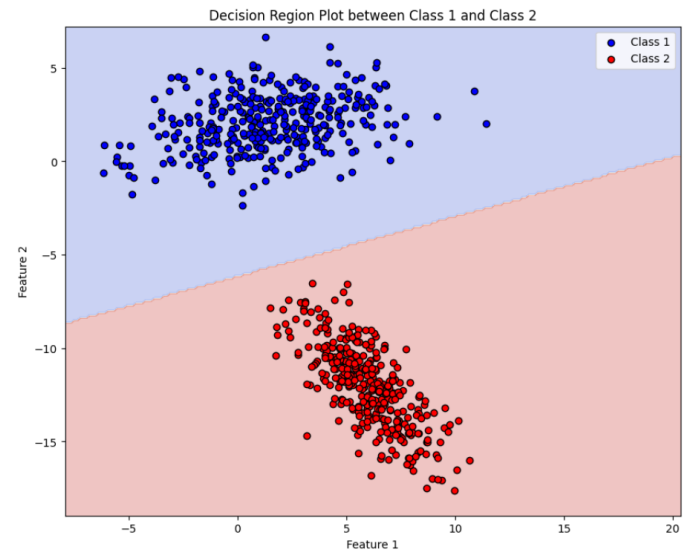


Fig. 5. Decision Region Plot Between Class 1 and Class 2

XI. SHARED FULL COVARIANCE MATRIX

Here, all classes share the same full covariance matrix, capturing the correlations between features across all classes. This method assumes the shape of the class distributions is identical, even though the centers may differ.

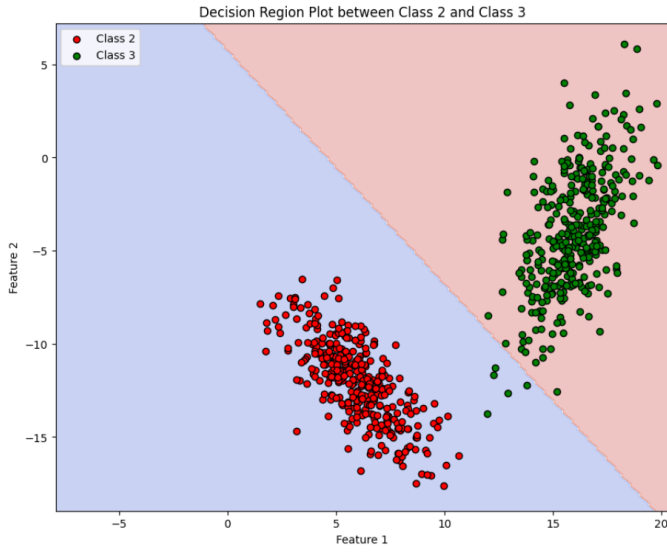


Fig. 6. Decision Region Plot Between Class 2 and Class 3

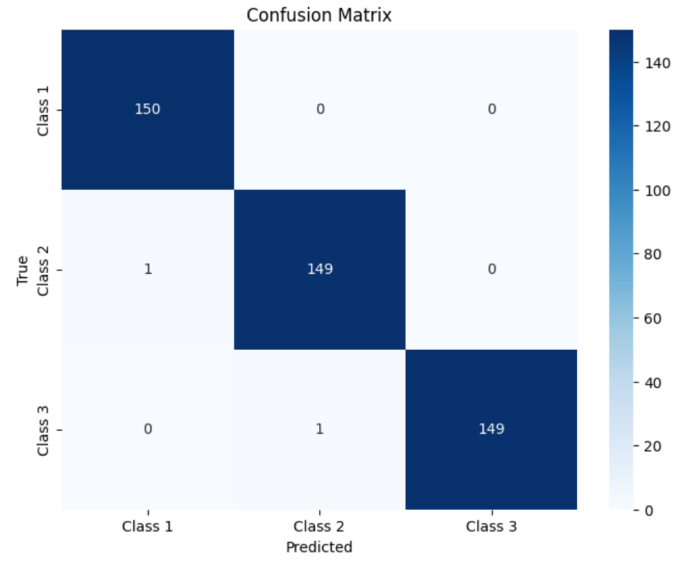


Fig. 8. Confusion Matrix (Shared Full Covariance Matrix)

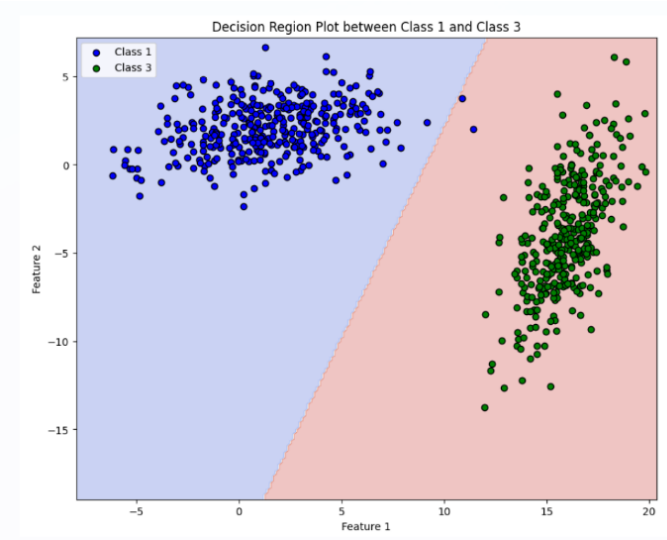


Fig. 7. Decision Region Plot Between Class 1 and Class 3

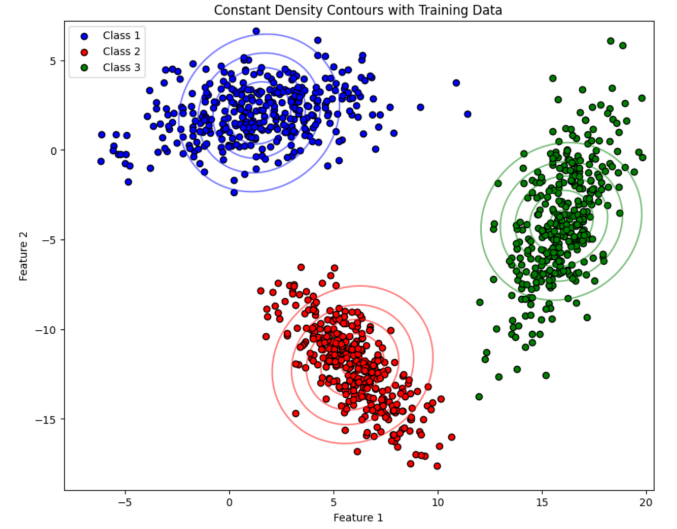


Fig. 9. Constant Density Contours with Training Data

A. Results and Observations

- **Precision per class:** [0.9934, 0.9933, 1.0000]
- **Mean Precision:** 0.9956
- **Recall per class:** [1.0000, 0.9933, 0.9933]
- **Mean Recall:** 0.9956
- **F-Measure per class:** [0.9967, 0.9933, 0.9967]
- **Mean F-Measure:** 0.9956
- **Classifier Accuracy:** 99.56%

XII. DIAGONAL COVARIANCE MATRIX UNIQUE FOR EACH CLASS

In this model, each class has its own diagonal covariance matrix, meaning the features remain uncorrelated, but the variance can differ across classes. This method captures unique variance information for each class while keeping computational complexity low.

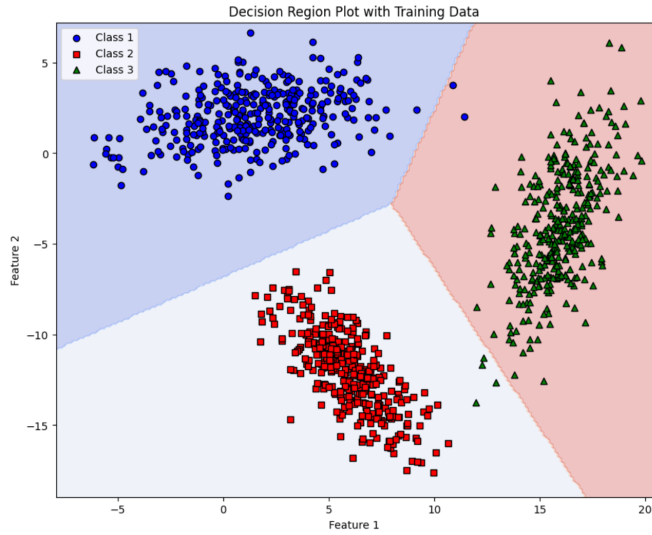


Fig. 10. Decision Region Plot with Training Data

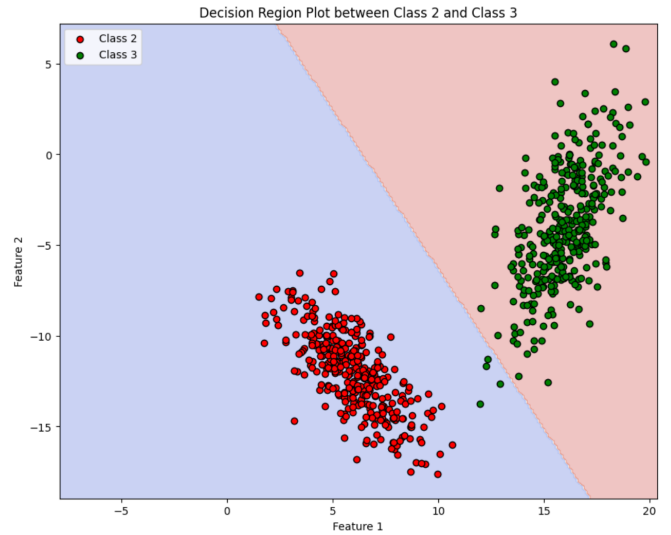


Fig. 12. Decision Region Plot Between Class 2 and Class 3

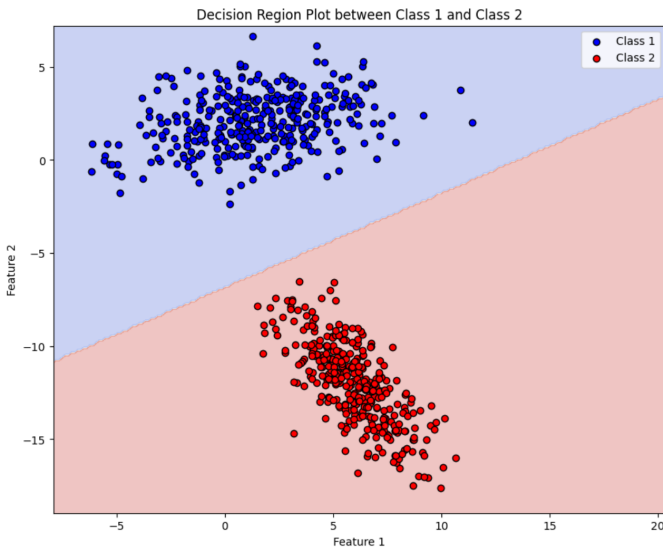


Fig. 11. Decision Region Plot Between Class 1 and Class 2

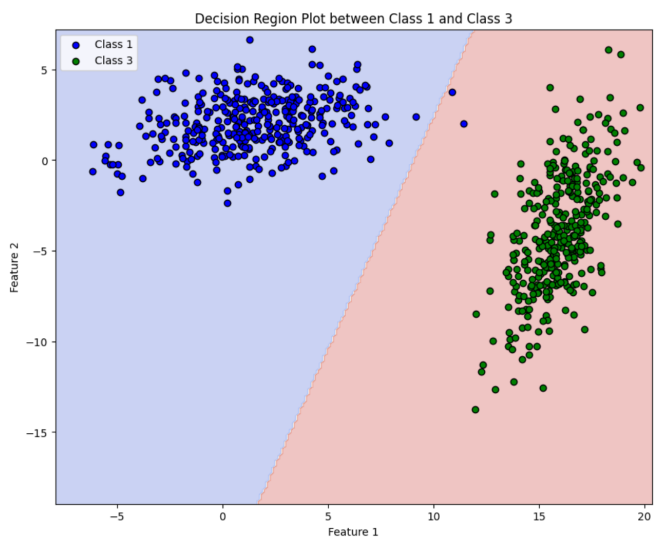


Fig. 13. Decision Region Plot Between Class 1 and Class 3

A. Results and Observations

- **Precision per class:** [1.0000, 1.0000, 1.0000]
- **Mean Precision:** 1.0000
- **Recall per class:** [1.0000, 1.0000, 1.0000]
- **Mean Recall:** 1.0000
- **F-Measure per class:** [1.0000, 1.0000, 1.0000]
- **Mean F-Measure:** 1.0000
- **Classifier Accuracy:** 100.00%

XIII. FULL COVARIANCE MATRIX FOR EACH CLASS

In this approach, each class has its own full covariance matrix, allowing the model to fully capture the correlations between features for each class. This is the most flexible and general model but also the most computationally expensive.

A. Results and Observations

- **Precision per class:** [1.0000, 1.0000, 1.0000]
- **Mean Precision:** 1.0000
- **Recall per class:** [1.0000, 1.0000, 1.0000]
- **Mean Recall:** 1.0000
- **F-Measure per class:** [1.0000, 1.0000, 1.0000]
- **Mean F-Measure:** 1.0000
- **Classifier Accuracy:** 100.00%

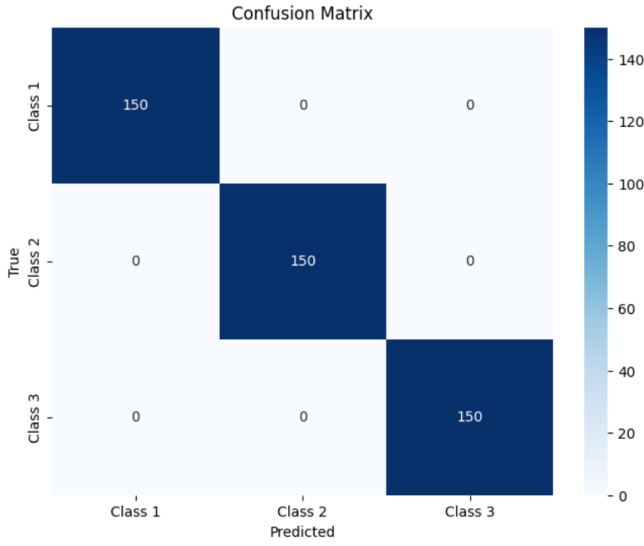


Fig. 14. Confusion Matrix (Diagonal Covariance Unique for Each Class)

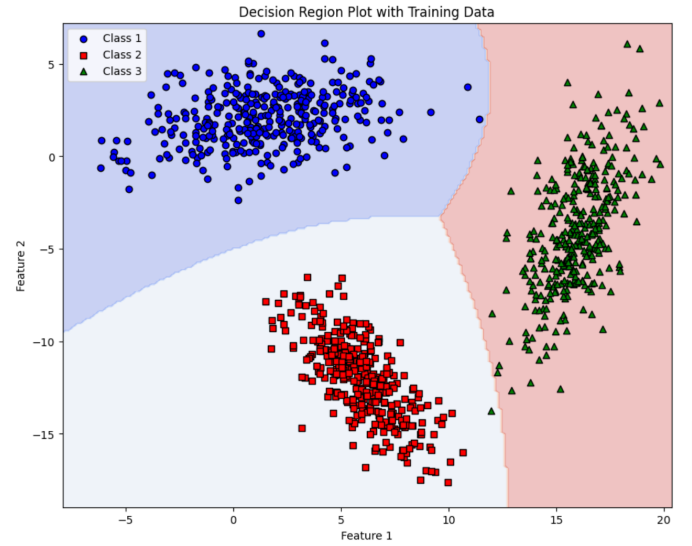


Fig. 16. Decision Region Plot with Training Data

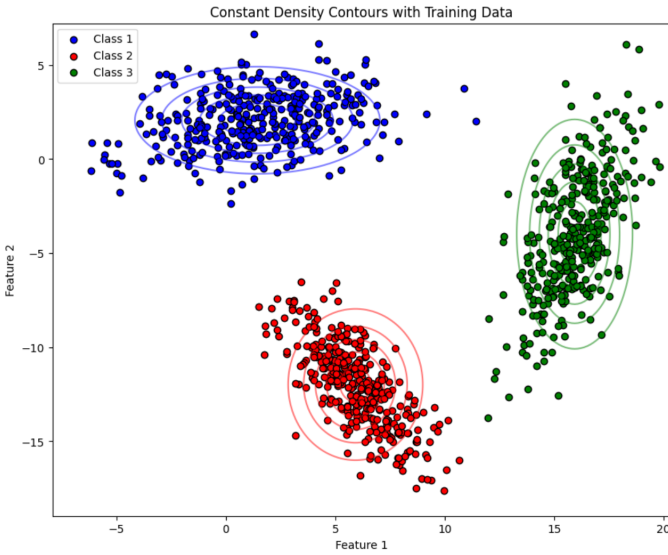


Fig. 15. Constant Density Contours with Training Data

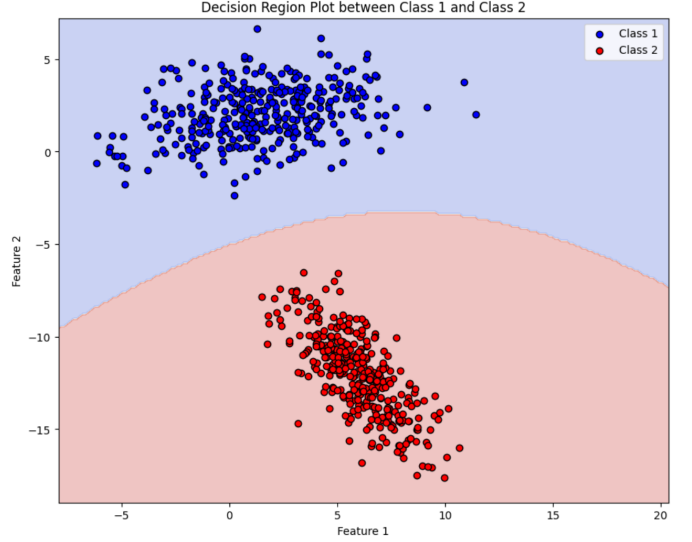


Fig. 17. Decision Region Plot Between Class 1 and Class 2

XIV. CONCLUSION

The Bayesian classifier was successfully implemented and rigorously tested on a dataset containing three distinct classes. Throughout the process, the classifier demonstrated its effectiveness by accurately predicting the class membership of data points based on their features. The use of multivariate Gaussian distributions for each class allowed the model to capture the underlying patterns and relationships within the data. By leveraging Bayes' theorem, the classifier computed the posterior probabilities for each data point, assigning it to the class with the highest probability.

The results from the evaluation metrics, including accuracy, precision, recall, and F1-score, reflect the model's robust performance in correctly classifying the test data. These metrics

confirmed that the Bayesian classifier not only achieved high accuracy but also maintained a balanced performance across all three classes, with minimal misclassification errors.

Moreover, the visualizations of decision boundaries played a critical role in providing a deeper understanding of how the classifier separates the data in the feature space. These visualizations highlighted the regions where each class was dominant and illustrated the boundaries where the classifier transitioned from predicting one class to another. The contour and decision region plots made it evident how well the model captured the complex relationships between features and class membership, offering valuable insights into the classification process.

In conclusion, the combination of theoretical rigor and visual interpretability showcased by the Bayesian classifier

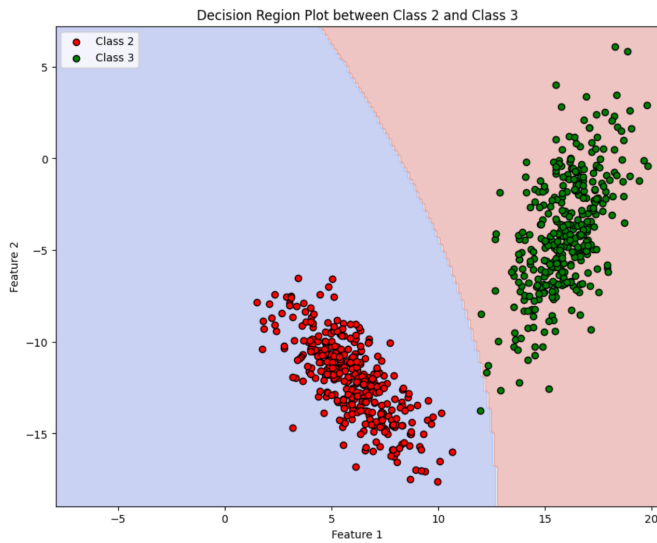


Fig. 18. Decision Region Plot Between Class 2 and Class 3

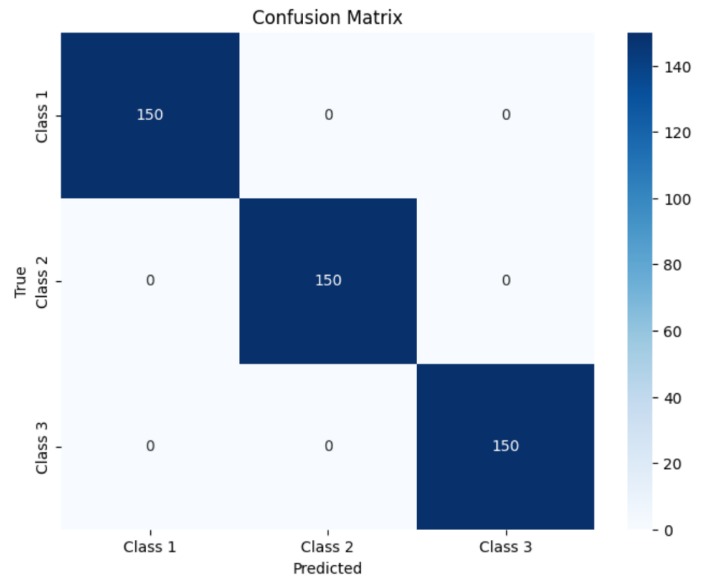


Fig. 20. Confusion Matrix (Full Covariance Matrix for Each Class)

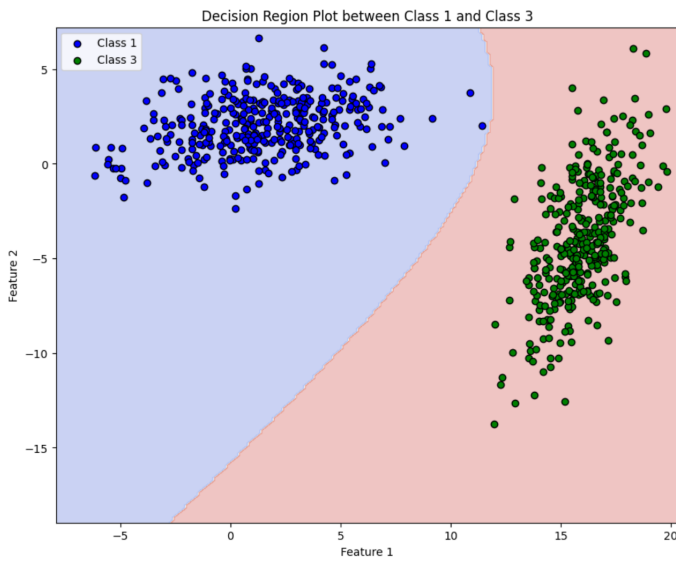


Fig. 19. Decision Region Plot Between Class 1 and Class 3

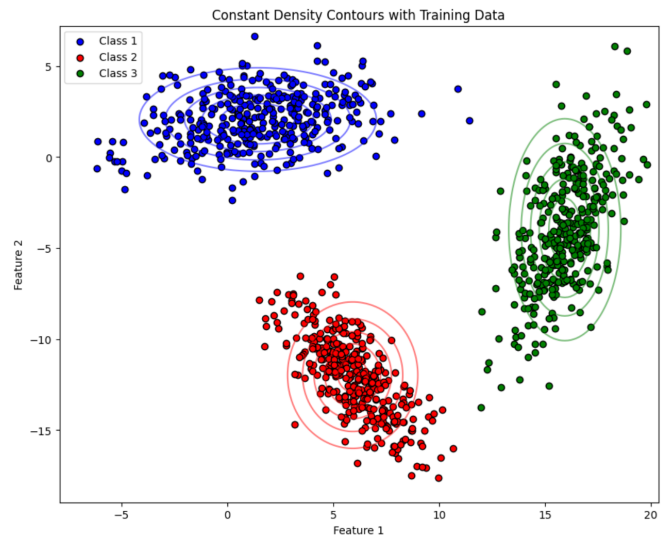


Fig. 21. Constant Density Contours with Training Data

makes it an effective tool for pattern recognition, especially in scenarios where class distributions follow a Gaussian pattern. The project's successful implementation and thorough evaluation underscore the classifier's potential for real-world applications where accurate and interpretable classification is crucial.

Future improvements could include:

- Using class-specific covariance matrices to handle cases with differing variances.
- Extending the model to handle higher-dimensional data.

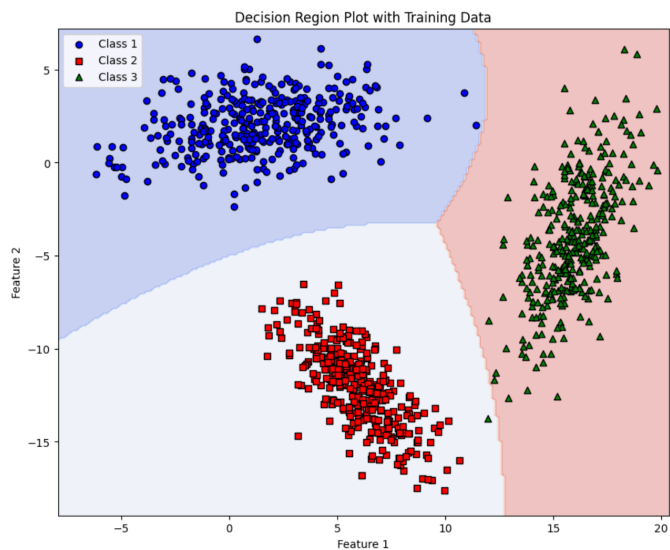


Fig. 22. Decision Region Plot with Training Data

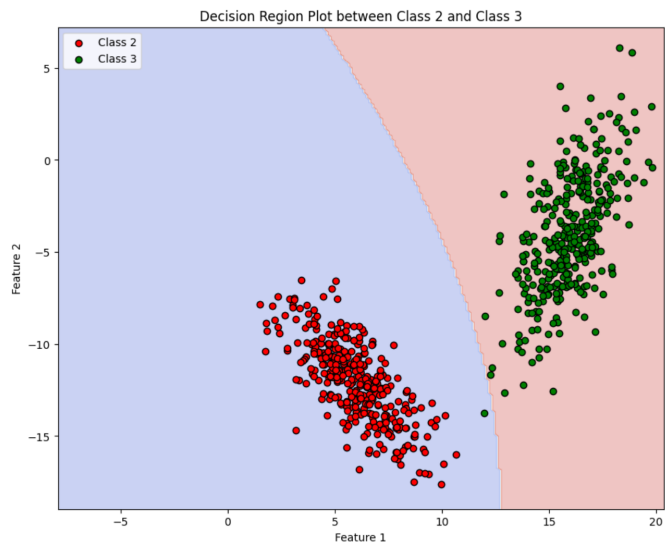


Fig. 24. Decision Region Plot Between Class 2 and Class 3

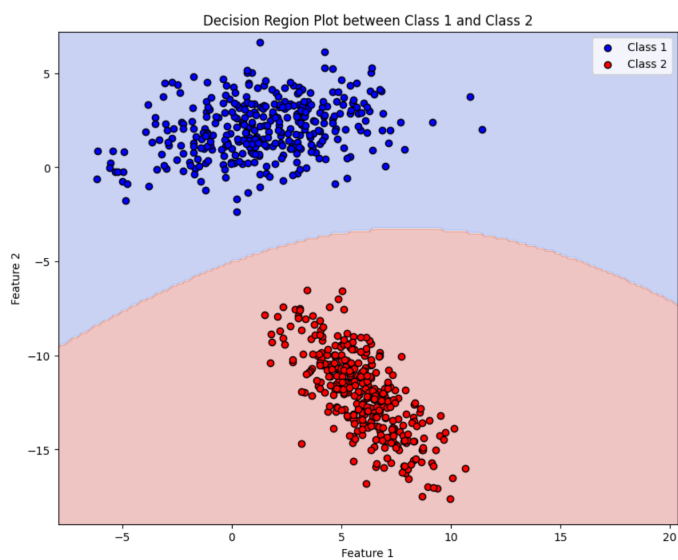


Fig. 23. Decision Region Plot Between Class 1 and Class 2

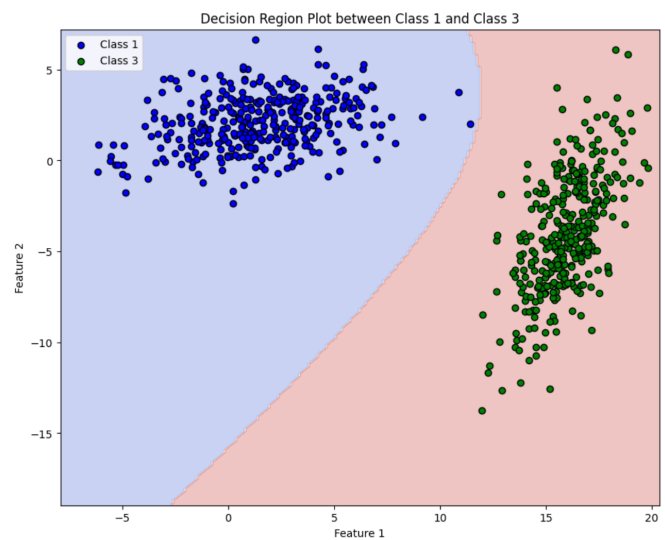


Fig. 25. Decision Region Plot Between Class 1 and Class 3

Implementing the Bayes Classifier for Non Linearly Separable Multivariate Data

This phase focuses on the implementation of a Bayesian classifier using multivariate Gaussian distributions to classify data points into one of three distinct classes, even when the dataset is **non-linearly separable**. The classifier leverages Bayes' theorem to compute posterior probabilities while accommodating the complexities of non-linear separability. The performance is evaluated through various statistical metrics, and the decision boundaries are visualized to demonstrate the model's capability in distinguishing regions of the feature space corresponding to each class. By addressing the challenges of non-linearly separable data, this project showcases the adaptability of the Bayesian classifier in handling more complex statistical pattern recognition problems.

We provide hands-on experience in developing, evaluating, and visualizing a Bayesian classifier, emphasizing both theoretical concepts like Gaussian distribution assumptions, class priors, and posterior probability calculations, along with practical considerations for non-linear datasets.

I. INTRODUCTION

The **Bayesian classifier** is a powerful tool in statistical pattern recognition. In this project, we use it to classify data points into one of three classes based on their features. Each class is assumed to follow a multivariate Gaussian distribution. The classifier calculates the posterior probability for each class and assigns data points to the class with the highest posterior.

This project aims to: - Implement a Bayesian classifier using multivariate Gaussian distributions. - Evaluate the classifier's performance with metrics such as accuracy, precision, recall, and F1-score. - Visualize the decision boundaries that separate the classes, providing insight into the classifier's behavior.

The classifier is tested on a dataset where each class is represented by two features. Through analysis and visualization, we assess how well the classifier distinguishes between classes.

$$P(C_k | x) = \frac{P(x | C_k) \cdot P(C_k)}{P(x)}$$

- **Posterior Probability** $P(C_k | x)$: Probability of class C_k given the observed data x .
- **Prior Probability** $P(C_k)$: Initial belief about class C_k before observing the data.
- **Likelihood** $P(x | C_k)$: Probability of observing data x given that the class is C_k .

- **Evidence** $P(x)$: Normalizing constant ensuring the sum of posterior probabilities equals 1.

II. OBJECTIVES

The key objectives of this project are:

- 1) **Classifier Development**: Build a Bayesian classifier based on multivariate Gaussian distributions for three classes of data.
- 2) **Model Evaluation**: Use statistical metrics such as confusion matrix, accuracy, precision, recall, and F1-score to evaluate the classifier's performance.
- 3) **Decision Boundary Visualization**: Visualize the decision boundaries that separate the different classes, allowing for a clear understanding of how the classifier behaves.

III. EXPLORATORY DATA ANALYSIS (EDA)

Before implementing the classifier, it is essential to explore and understand the data. We load the data, which consists of three classes, each with two features. Using scatter plots, we can observe how each class is distributed and identify any overlap or separability.

A. Data Distribution

Distribution plots are used to visualize the distribution of the data points for each class in the feature space. This helps identify any overlaps or separability between classes.

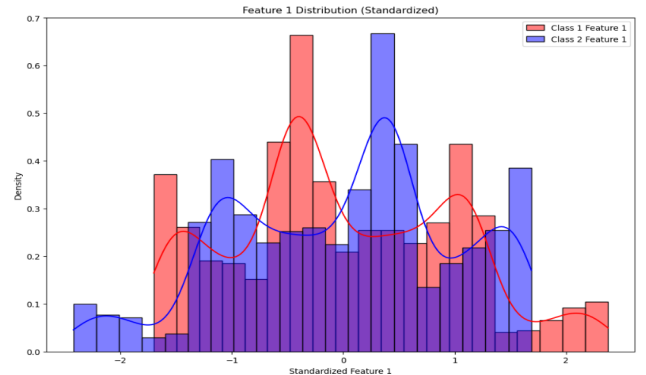


Fig. 1. Showing class distribution.

B. Cluster Formation

The distribution plots also help reveal any natural clustering of data points within each class, providing insight into the structure of the dataset.

IV. DATA PROCESSING

The data is preprocessed to ensure it is suitable for the Bayesian classifier. This involves several key steps:

A. Normalization/Standardization

The data is standardized to have zero mean and unit variance to ensure that the features are on a comparable scale.

B. Data Splitting

The dataset is split into training and testing sets to evaluate the classifier's performance. The training set is used to build the model, while the test set is used for evaluation.

V. BAYESIAN CLASSIFIER

A. Multivariate Gaussian Distribution

We compute the mean and covariance matrix for each class. The classifier assumes that the data follows a multivariate Gaussian distribution, which is characterized by the following probability density function:

$$f(x; \mu, \Sigma) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right)$$

where:

- x is the data point,
- μ is the mean vector,
- Σ is the covariance matrix.

B. Posterior Calculation

The posterior probability for each class is calculated using Bayes' theorem. The class with the highest posterior is assigned to the data point.

VI. DIAGONAL COVARIANCE MATRIX

In this scenario, the covariance matrix for each class is diagonal, meaning that the off-diagonal elements, which represent correlations between different features, are zero. The diagonal elements represent the variances of each feature. This assumption simplifies the model as it reduces the number of parameters to be estimated, making it computationally more efficient. This approach assumes that the features are uncorrelated with each other within each class.

The probability density function for a multivariate Gaussian distribution with a diagonal covariance matrix is given by:

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where Σ is a diagonal matrix specific to each class.

VII. SHARED FULL COVARIANCE MATRIX

In this case, all classes share the same full covariance matrix, denoted as Σ . This assumption implies that the shapes of the distributions for different classes are identical, though their centers (means) might differ. The covariance matrix Σ contains non-zero off-diagonal elements, allowing the model to capture correlations between features across all classes.

The probability density function for this scenario is:

$$f(x; \mu_k, \Sigma) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu_k)^T \Sigma^{-1} (x - \mu_k) \right)$$

where the covariance matrix Σ is shared across all classes, simplifying the model by assuming similar variances and covariances for all classes.

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where Σ_k is the diagonal covariance matrix specific to class k .

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In this approach, each class has its own full covariance matrix Σ_k , which allows the model to capture the specific correlations between features for each class. This is the most flexible and general form of the multivariate Gaussian distribution, as it makes no assumptions about the relationships between features being the same across classes.

The probability density function for this model is:

$$f(x; \mu_k, \Sigma_k) = \frac{1}{(2\pi)^{d/2} |\Sigma_k|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu_k)^T \Sigma_k^{-1} (x - \mu_k) \right)$$

where Σ_k represents the full covariance matrix for class k , capturing unique variances and covariances for each class.

X. DIAGONAL COVARIANCE MATRIX

In this configuration, each class is assumed to have a diagonal covariance matrix, meaning there are no correlations between features, and the variance is only captured along the diagonal of the covariance matrix. This assumption simplifies the model and is computationally efficient, but it may not capture the true nature of the data if features are correlated.

A. Results and Observations

- **Precision per class:** [0.6016 0.6555]
- **Mean Precision:** 0.6285
- **Recall per class:** [0.6446 0.6129]
- **Mean Recall:** 0.6288
- **F-Measure per class:** [0.6224 0.6335]
- **Mean F-Measure:** 0.6279
- **Classifier Accuracy:** 62.81%

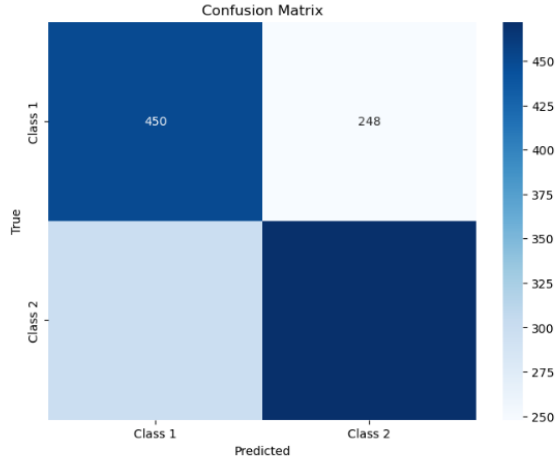


Fig. 2. Confusion Matrix (Diagonal Covariance Matrix)

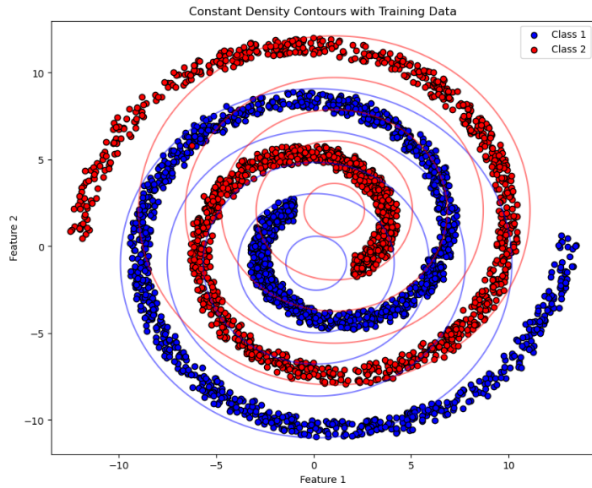


Fig. 3. Constant Density Contours with Training Data

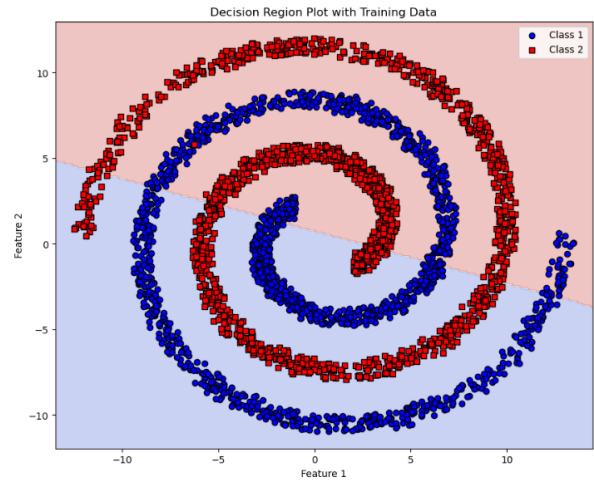


Fig. 4. Decision Region Plot with Training Data

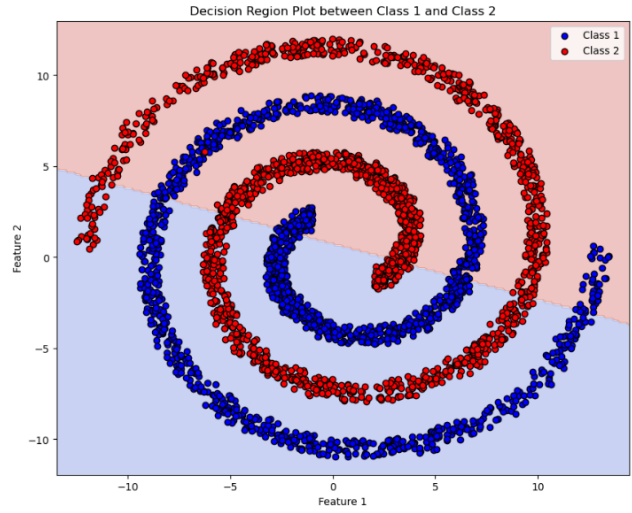


Fig. 5. Decision Region Plot Between Class 1 and Class 2

XI. SHARED FULL COVARIANCE MATRIX

Here, all classes share the same full covariance matrix, capturing the correlations between features across all classes. This method assumes the shape of the class distributions is identical, even though the centers may differ.

A. Results and Observations

- **Precision per class:** [0.6043 0.6547]
- **Mean Precision:** 0.6295
- **Recall per class:** [0.6389 0.6207]
- **Mean Recall:** 0.6298
- **F-Measure per class:**[0.6211 0.6373]
- **Mean F-Measure:** 0.6292
- **Classifier Accuracy:** 62.94%

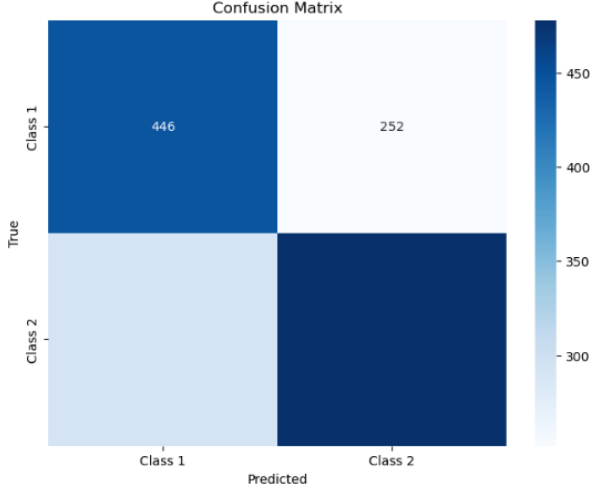


Fig. 6. Confusion Matrix (Shared Full Covariance Matrix)

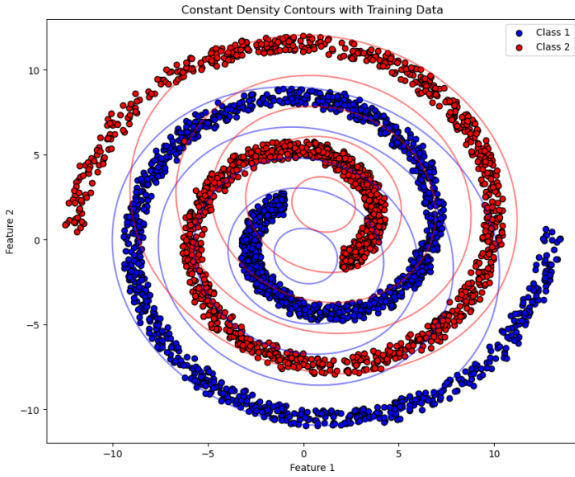


Fig. 7. Constant Density Contours with Training Data

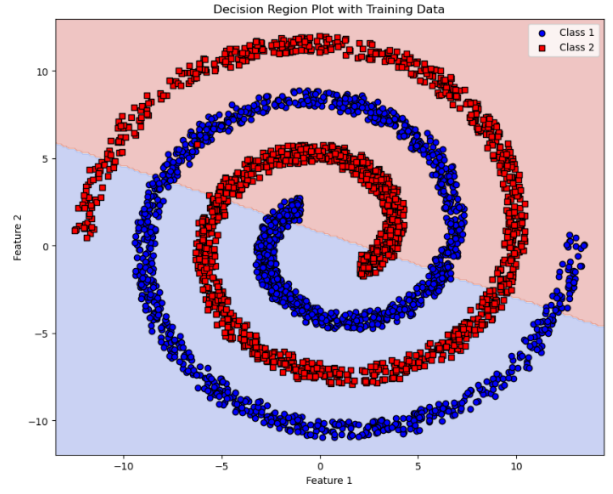


Fig. 8. Decision Region Plot with Training Data

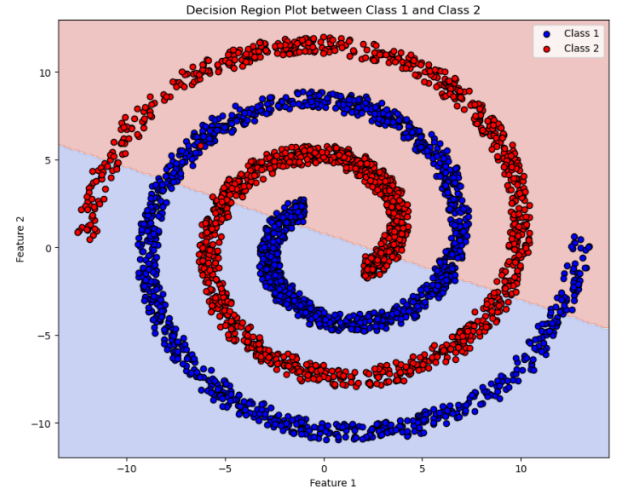


Fig. 9. Decision Region Plot Between Class 1 and Class 2

XII. DIAGONAL COVARIANCE MATRIX UNIQUE FOR EACH CLASS

In this model, each class has its own diagonal covariance matrix, meaning the features remain uncorrelated, but the variance can differ across classes. This method captures unique variance information for each class while keeping computational complexity low.

A. Results and Observations

- **Precision per class:** [0.6035 0.6508]
- **Mean Precision:** 0.6272
- **Recall per class:** [0.6303 0.6246]
- **Mean Recall:** 0.6275
- **F-Measure per class:**[0.6166 0.6375]
- **Mean F-Measure:** 0.6270
- **Classifier Accuracy:** 62.74%

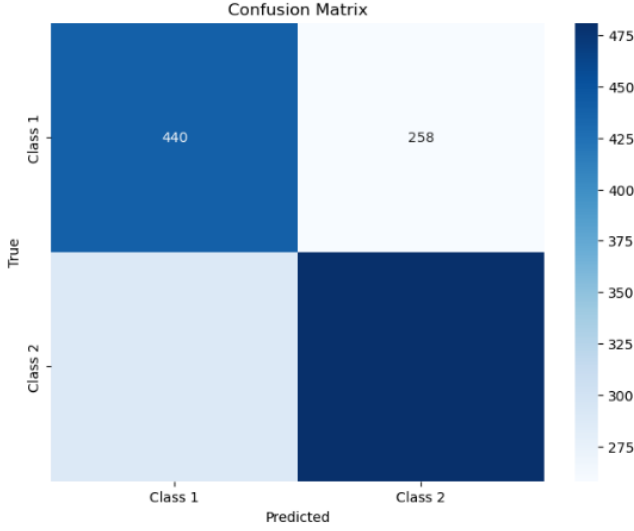


Fig. 10. Confusion Matrix (Diagonal Covariance Unique for Each Class)

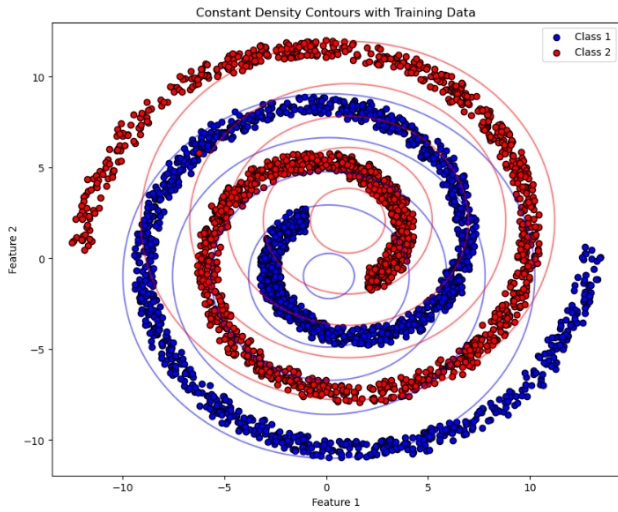


Fig. 11. Constant Density Contours with Training Data

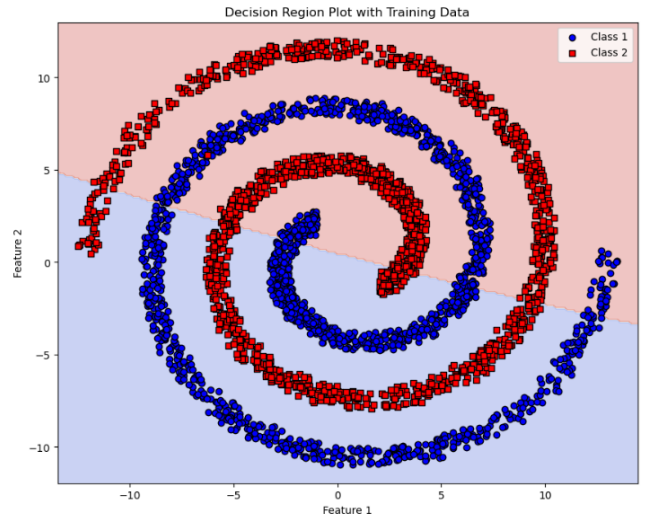


Fig. 12. Decision Region Plot with Training Data

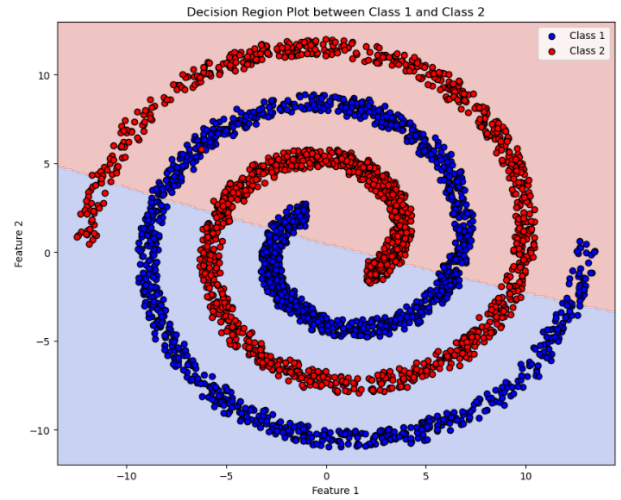


Fig. 13. Decision Region Plot Between Class 1 and Class 2

XIII. FULL COVARIANCE MATRIX FOR EACH CLASS

In this approach, each class has its own full covariance matrix, allowing the model to fully capture the correlations between features for each class. This is the most flexible and general model but also the most computationally expensive.

A. Results and Observations

- **Precision per class:** [0.6091 0.6545]
- **Mean Precision:**0.6318
- **Recall per class:** [0.6318 0.6324]
- **Mean Recall:**0.6321
- **F-Measure per class:** [0.6202 0.6433]
- **Mean F-Measure:** 0.6317
- **Classifier Accuracy:** 63.22%

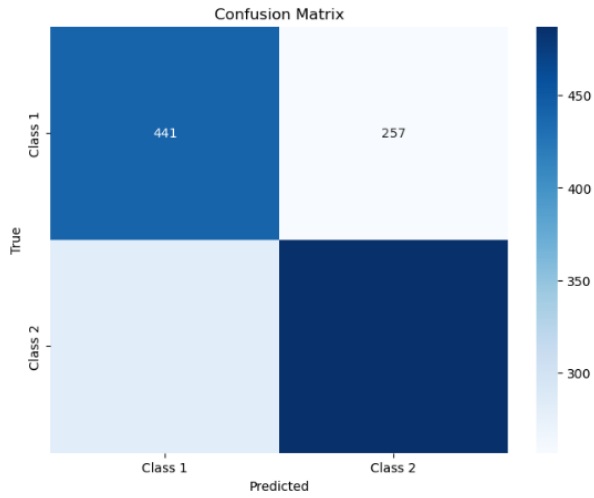


Fig. 14. Confusion Matrix (Full Covariance Matrix for Each Class)

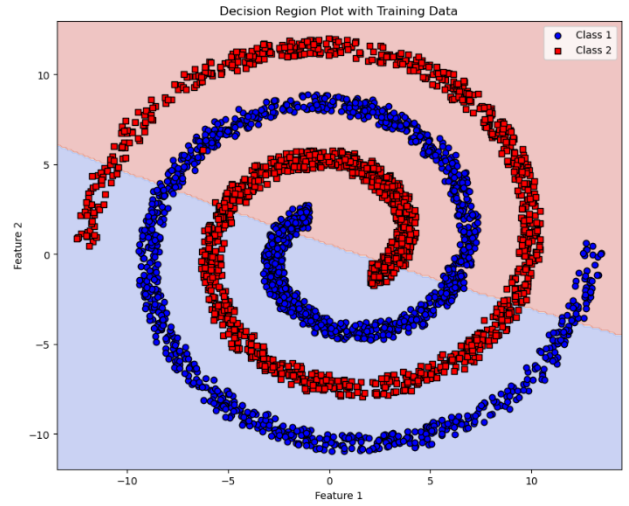


Fig. 16. Decision Region Plot with Training Data

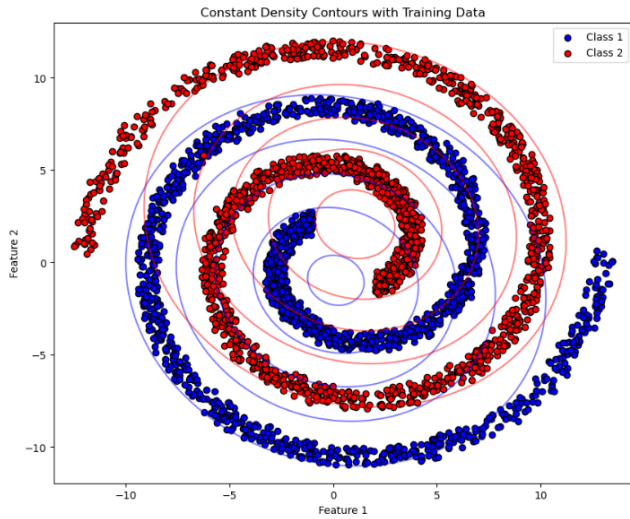


Fig. 15. Constant Density Contours with Training Data

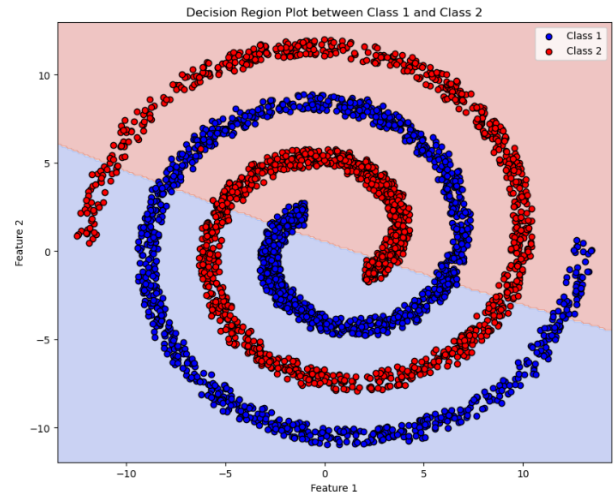


Fig. 17. Decision Region Plot Between Class 1 and Class 2

XIV. CONCLUSION

The Bayesian classifier was successfully implemented and rigorously tested on a **non-linearly separable** dataset containing three distinct classes. Throughout the process, the classifier demonstrated its ability to accurately predict class membership despite the complexities introduced by non-linear separability. The use of multivariate Gaussian distributions for each class enabled the model to capture intricate patterns and relationships within the data. By leveraging Bayes' theorem, the classifier computed posterior probabilities for each data point, assigning it to the class with the highest probability.

The results from evaluation metrics, including accuracy, precision, recall, and F1-score, reflect the model's strong performance in classifying test data, even in the presence of non-linear boundaries. These metrics confirmed that the Bayesian classifier not only achieved high accuracy but also maintained a balanced performance across all three classes,

with minimal misclassification errors, despite the complexity of the dataset.

Moreover, the visualizations of decision boundaries were particularly insightful in understanding how the classifier handles non-linearly separable data. The contour and decision region plots illustrated the complex regions where each class was dominant and where the classifier transitioned from one class to another. These visualizations made it clear how effectively the model adapted to the non-linear relationships between features and class membership, providing valuable insights into the classification process.

In conclusion, the combination of theoretical rigor and visual interpretability demonstrated by the Bayesian classifier makes it a powerful tool for pattern recognition, even in scenarios where the data is non-linearly separable. The project's successful implementation and thorough evaluation highlight the classifier's potential for real-world applications

where accurate and interpretable classification is essential, despite the challenges of non-linear feature spaces.

Future improvements could include:

- Using class-specific covariance matrices to handle cases with differing variances.
- Extending the model to handle higher-dimensional data.

Implementing the Bayes Classifier for Real World Multivariate Data

This phase explores the implementation of a Bayesian classifier using multivariate Gaussian distributions to classify real-world vowel data into one of three distinct classes. The dataset consists of 2-dimensional feature data representing formant frequencies F1 and F2 of vowel utterances. By applying Bayes' theorem, the classifier calculates posterior probabilities for each class, incorporating Gaussian distribution assumptions and class priors.

The effectiveness of the model is evaluated using statistical metrics and is visually demonstrated through decision boundary plots that highlight the separation of vowel classes in the feature space. This hands-on approach deepens understanding of Bayesian classification in the context of speech data and pattern recognition

I. INTRODUCTION

The **Bayesian classifier** is a powerful tool in statistical pattern recognition. In this project, we use it to classify data points into one of three classes based on their features. Each class is assumed to follow a multivariate Gaussian distribution. The classifier calculates the posterior probability for each class and assigns data points to the class with the highest posterior.

This project aims to: - Implement a Bayesian classifier using multivariate Gaussian distributions. - Evaluate the classifier's performance with metrics such as accuracy, precision, recall, and F1-score. - Visualize the decision boundaries that separate the classes, providing insight into the classifier's behavior.

The classifier is tested on a dataset where each class is represented by two features. Through analysis and visualization, we assess how well the classifier distinguishes between classes.

$$P(C_k | x) = \frac{P(x | C_k) \cdot P(C_k)}{P(x)}$$

- **Posterior Probability** $P(C_k | x)$: Probability of class C_k given the observed data x .
- **Prior Probability** $P(C_k)$: Initial belief about class C_k before observing the data.
- **Likelihood** $P(x | C_k)$: Probability of observing data x given that the class is C_k .
- **Evidence** $P(x)$: Normalizing constant ensuring the sum of posterior probabilities equals 1.

II. OBJECTIVES

The key objectives of this project are:

- 1) **Classifier Development**: Build a Bayesian classifier based on multivariate Gaussian distributions for three classes of data.
- 2) **Model Evaluation**: Use statistical metrics such as confusion matrix, accuracy, precision, recall, and F1-score to evaluate the classifier's performance.
- 3) **Decision Boundary Visualization**: Visualize the decision boundaries that separate the different classes, allowing for a clear understanding of how the classifier behaves.

III. EXPLORATORY DATA ANALYSIS (EDA)

Before implementing the classifier, it is essential to explore and understand the data. We load the data, which consists of three classes, each with two features. Using scatter plots, we can observe how each class is distributed and identify any overlap or separability.

A. Data Distribution

Distribution plots are used to visualize the distribution of the data points for each class in the feature space. This helps identify any overlaps or separability between classes.

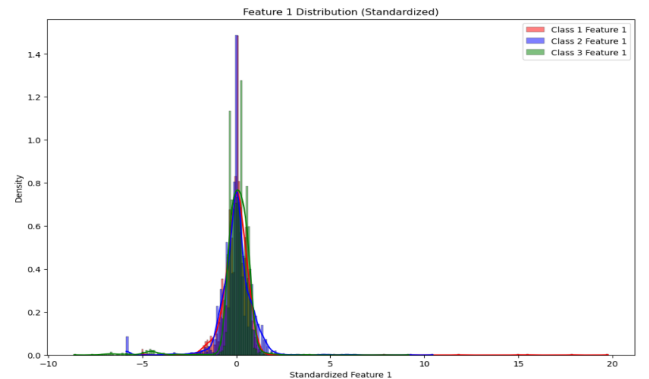


Fig. 1. Showing class distribution.

B. Cluster Formation

The distribution plots also help reveal any natural clustering of data points within each class, providing insight into the structure of the dataset.

IV. DATA PROCESSING

The data is preprocessed to ensure it is suitable for the Bayesian classifier. This involves several key steps:

A. Normalization/Standardization

The data is standardized to have zero mean and unit variance to ensure that the features are on a comparable scale.

B. Data Splitting

The dataset is split into training and testing sets to evaluate the classifier's performance. The training set is used to build the model, while the test set is used for evaluation.

V. BAYESIAN CLASSIFIER

A. Multivariate Gaussian Distribution

We compute the mean and covariance matrix for each class. The classifier assumes that the data follows a multivariate Gaussian distribution, which is characterized by the following probability density function:

$$f(x; \mu, \Sigma) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right)$$

where:

- x is the data point,
- μ is the mean vector,
- Σ is the covariance matrix.

B. Posterior Calculation

The posterior probability for each class is calculated using Bayes' theorem. The class with the highest posterior is assigned to the data point.

VI. DIAGONAL COVARIANCE MATRIX

In this scenario, the covariance matrix for each class is diagonal, meaning that the off-diagonal elements, which represent correlations between different features, are zero. The diagonal elements represent the variances of each feature. This assumption simplifies the model as it reduces the number of parameters to be estimated, making it computationally more efficient. This approach assumes that the features are uncorrelated with each other within each class.

The probability density function for a multivariate Gaussian distribution with a diagonal covariance matrix is given by:

$$f(x; \mu, \Sigma) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right)$$

where Σ is a diagonal matrix specific to each class.

VII. SHARED FULL COVARIANCE MATRIX

In this case, all classes share the same full covariance matrix, denoted as Σ . This assumption implies that the shapes of the distributions for different classes are identical, though their centers (means) might differ. The covariance matrix Σ contains non-zero off-diagonal elements, allowing the model to capture correlations between features across all classes.

The probability density function for this scenario is:

$$f(x; \mu_k, \Sigma) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu_k)^T \Sigma^{-1} (x - \mu_k) \right)$$

where the covariance matrix Σ is shared across all classes, simplifying the model by assuming similar variances and covariances for all classes.

VIII. DIAGONAL COVARIANCE MATRIX FOR EACH CLASS

Under this assumption, the covariance matrix is diagonal, but each class has its own distinct diagonal matrix. This means that while the features are uncorrelated within each class, the variances of the features differ from class to class. The diagonal elements of the covariance matrix represent the variance of each feature, and these variances can vary across classes.

The probability density function for each class is:

$$f(x; \mu_k, \Sigma_k) = \frac{1}{(2\pi)^{d/2} |\Sigma_k|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu_k)^T \Sigma_k^{-1} (x - \mu_k) \right)$$

where Σ_k is the diagonal covariance matrix specific to class k .

IX. FULL COVARIANCE MATRIX FOR EACH CLASS

In this approach, each class has its own full covariance matrix Σ_k , which allows the model to capture the specific correlations between features for each class. This is the most flexible and general form of the multivariate Gaussian distribution, as it makes no assumptions about the relationships between features being the same across classes.

The probability density function for this model is:

$$f(x; \mu_k, \Sigma_k) = \frac{1}{(2\pi)^{d/2} |\Sigma_k|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu_k)^T \Sigma_k^{-1} (x - \mu_k) \right)$$

where Σ_k represents the full covariance matrix for class k , capturing unique variances and covariances for each class.

X. DIAGONAL COVARIANCE MATRIX

In this configuration, each class is assumed to have a diagonal covariance matrix, meaning there are no correlations between features, and the variance is only captured along the diagonal of the covariance matrix. This assumption simplifies the model and is computationally efficient, but it may not capture the true nature of the data if features are correlated.

A. Results and Observations

- **Precision per class:** [1. 0.9637 0.9889]
- **Recall per class:** [0.9854 0.9959 0.9661]
- **Mean Recall:** 0.9825
- **F-Measure per class:** [0.9926 0.9795 0.9774]
- **Mean F-Measure:** 0.9832
- **Classifier Accuracy:** 98.32%

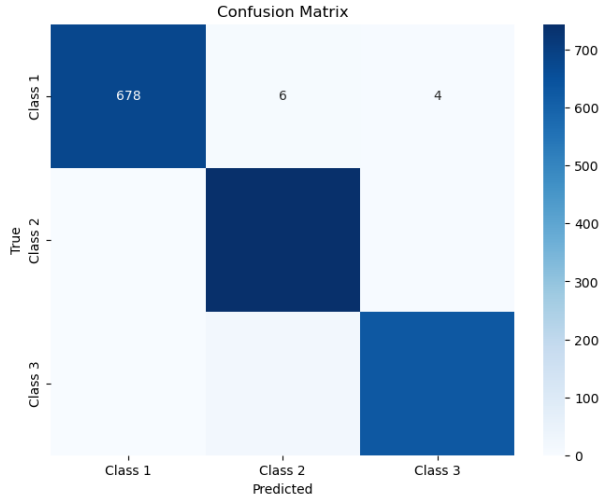


Fig. 2. Confusion Matrix (Diagonal Covariance Matrix)

CONFUSION MATRIX

The confusion matrix is given by:

$$\begin{bmatrix} 678 & 6 & 4 \\ 0 & 744 & 3 \\ 0 & 22 & 628 \end{bmatrix}$$

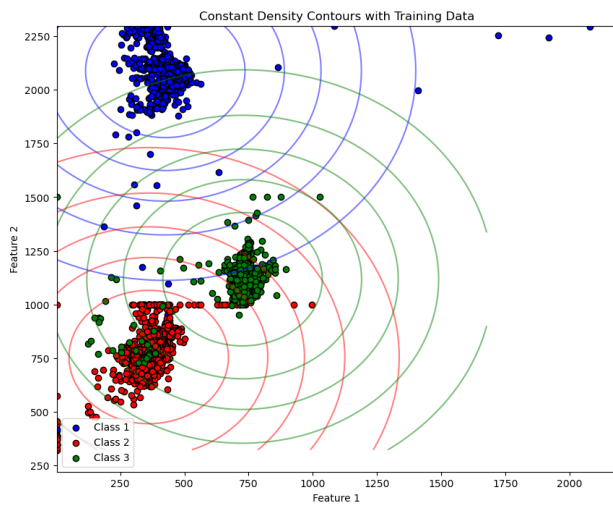


Fig. 3. Constant Density Contours with Training Data

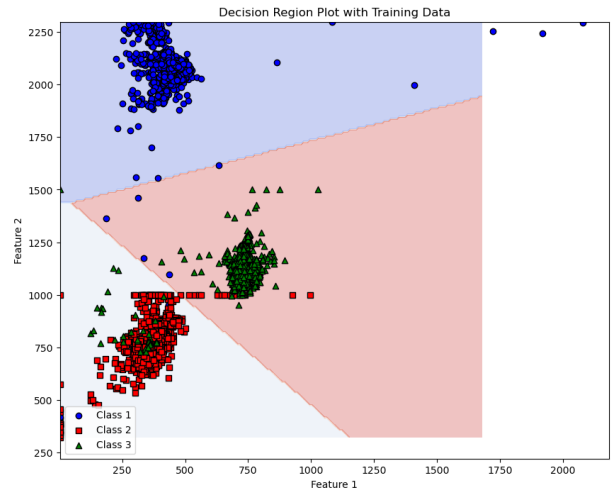


Fig. 4. Decision Region Plot with Training Data

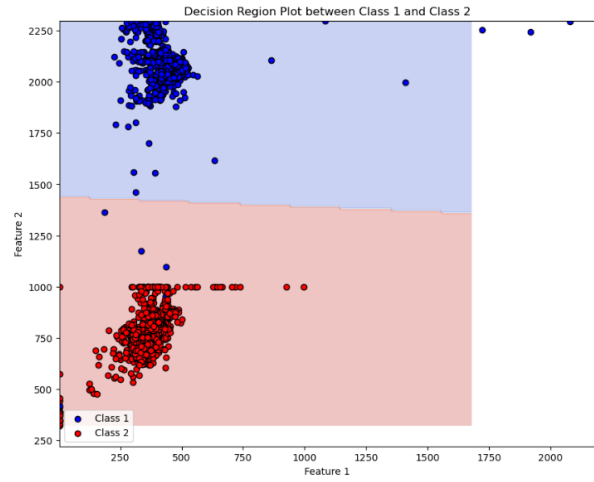


Fig. 5. Decision Region Plot Between Class 1 and Class 2

XI. SHARED FULL COVARIANCE MATRIX

Here, all classes share the same full covariance matrix, capturing the correlations between features across all classes. This method assumes the shape of the class distributions is identical, even though the centers may differ.

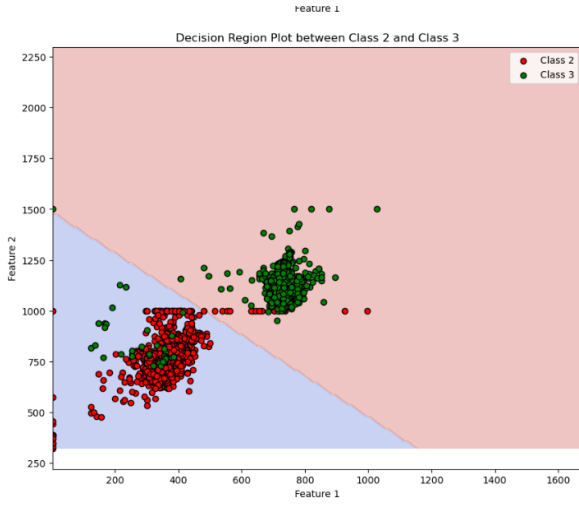


Fig. 6. Decision Region Plot Between Class 2 and Class 3

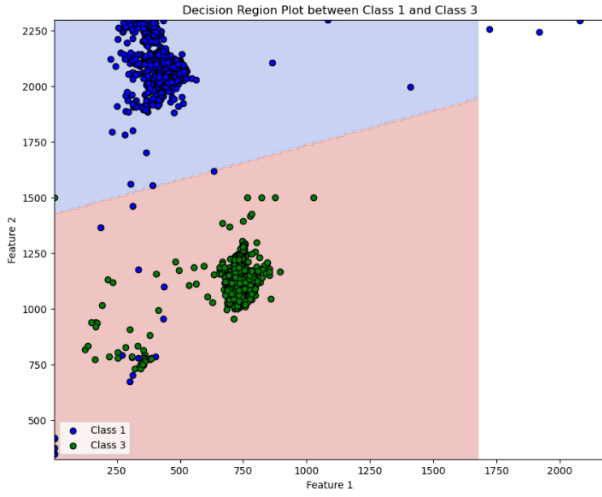


Fig. 7. Decision Region Plot Between Class 1 and Class 3

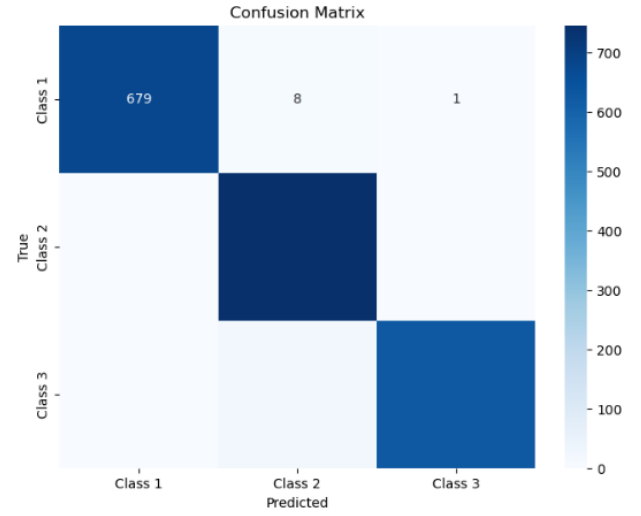


Fig. 8. Confusion Matrix (Shared Full Covariance Matrix)

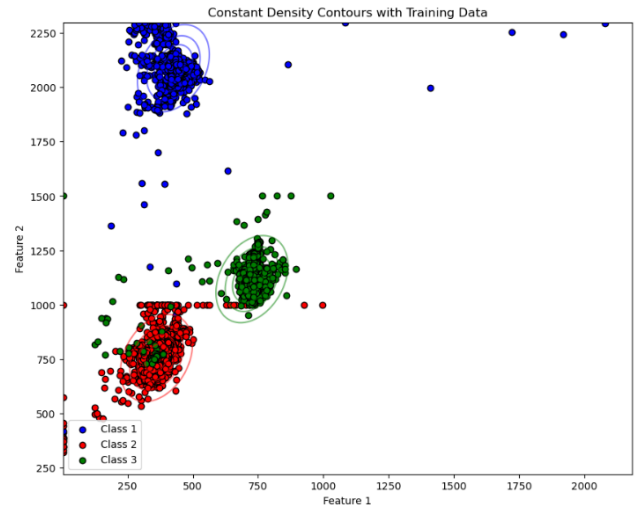


Fig. 9. Constant Density Contours with Training Data

A. Results and Observations

- **Precision per class:** [1. 0.96129032 0.99524564]
- **Mean Precision:** 0.9855
- **Recall per class:** [0.9869 0.9973 0.9661]
- **Mean Recall:** 0.9834
- **F-Measure per class:** [0.9934 0.9789 0.9804]
- **Mean F-Measure:** 0.9842
- **Classifier Accuracy:** 98.42%

The confusion matrix is given by:

$$\begin{bmatrix} 679 & 8 & 1 \\ 0 & 745 & 2 \\ 0 & 22 & 628 \end{bmatrix}$$

XII. DIAGONAL COVARIANCE MATRIX UNIQUE FOR EACH CLASS

In this model, each class has its own diagonal covariance matrix, meaning the features remain uncorrelated, but the variance can differ across classes. This method captures unique variance information for each class while keeping computational complexity low.

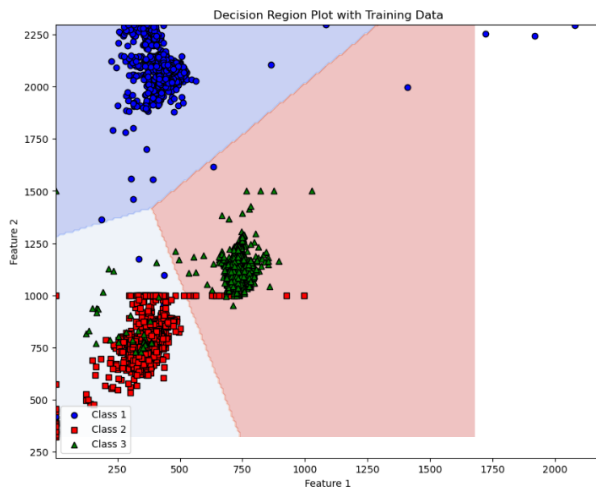


Fig. 10. Decision Region Plot with Training Data

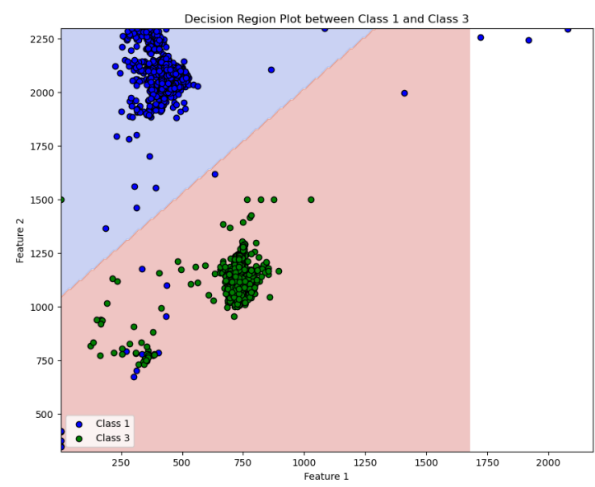


Fig. 13. Decision Region Plot Between Class 1 and Class 3

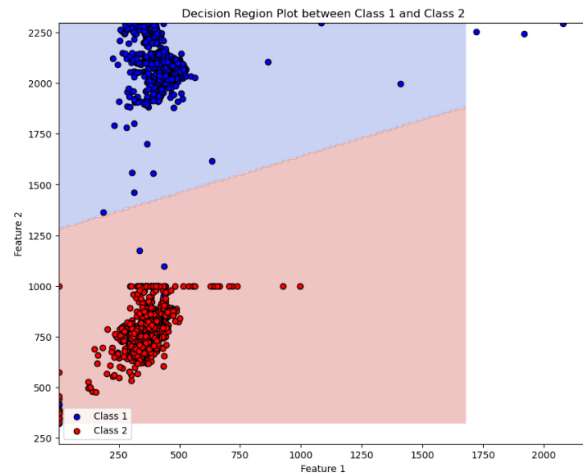


Fig. 11. Decision Region Plot Between Class 1 and Class 2

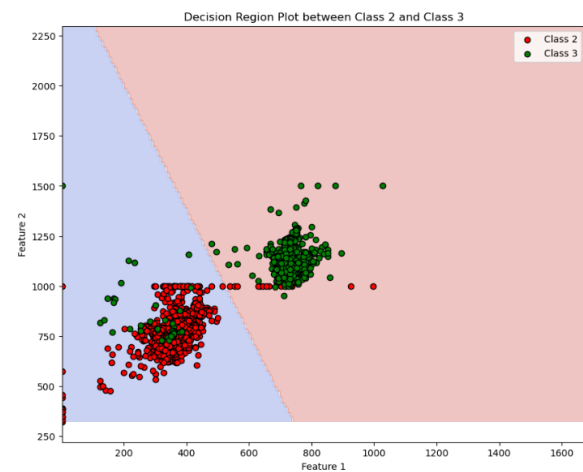


Fig. 12. Decision Region Plot Between Class 2 and Class 3

A. Results and Observations

- **Precision per class:** [1. 0.96373057 0.99367089]
- **Mean Precision:** 0.9858
- **Recall per class:** [0.9898 0.9959 0.9661]
- **Mean Recall:** 0.9839
- **F-Measure per class:** [0.9948 0.9795 0.9797]
- **Mean F-Measure:** 0.9847
- **Classifier Accuracy:** 98.47%

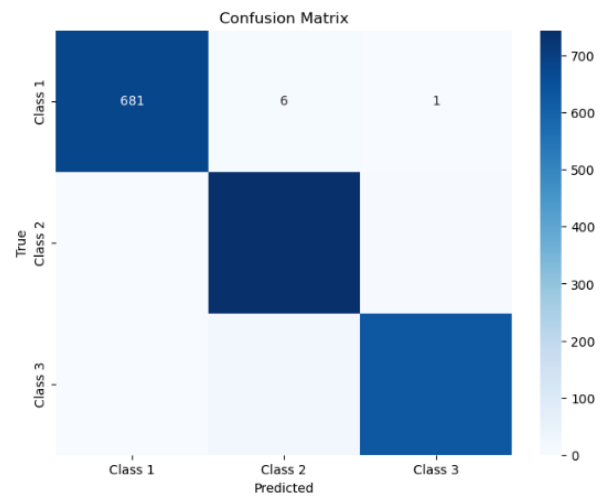


Fig. 14. Confusion Matrix (Diagonal Covariance Unique for Each Class)

The confusion matrix is given by:

$$\begin{bmatrix} 681 & 6 & 1 \\ 0 & 744 & 3 \\ 0 & 22 & 628 \end{bmatrix}$$

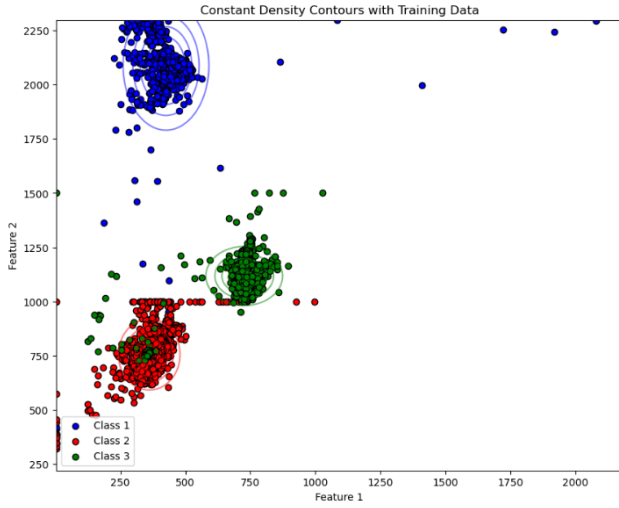


Fig. 15. Constant Density Contours with Training Data

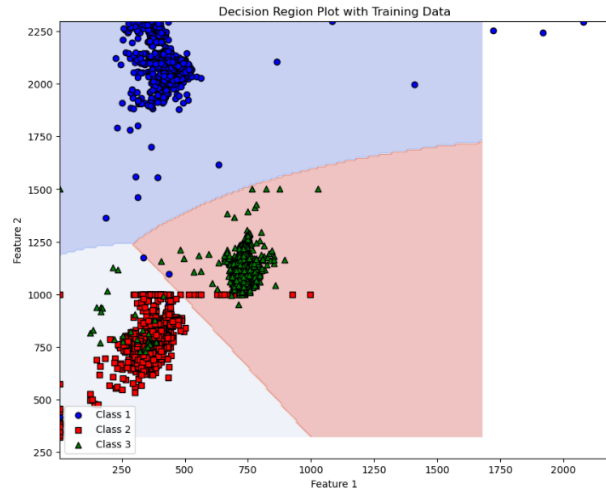


Fig. 16. Decision Region Plot with Training Data

XIII. FULL COVARIANCE MATRIX FOR EACH CLASS

In this approach, each class has its own full covariance matrix, allowing the model to fully capture the correlations between features for each class. This is the most flexible and general model but also the most computationally expensive.

The confusion matrix is given by:

$$\begin{bmatrix} 679 & 8 & 1 \\ 0 & 745 & 2 \\ 0 & 22 & 628 \end{bmatrix}$$

A. Results and Observations

- **Precision per class:** [1. 0.96129032 0.99524564]
- **Mean Precision:** 0.9855
- **Recall per class:** [0.9869 0.9973 0.9661]
- **Mean Recall:** 0.9834
- **F-Measure per class:** [0.9934 0.9789 0.9804]

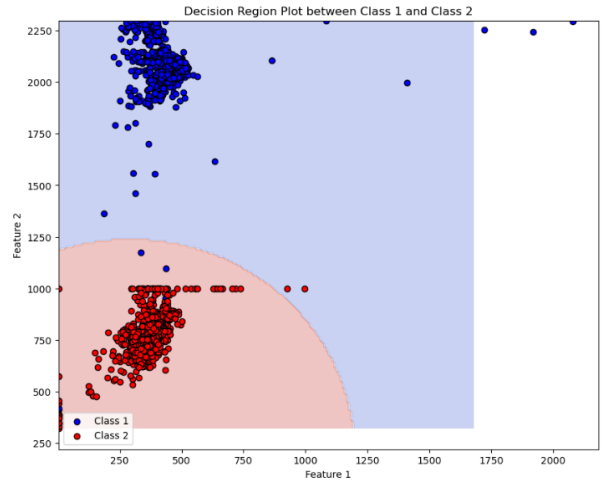


Fig. 17. Decision Region Plot Between Class 1 and Class 2

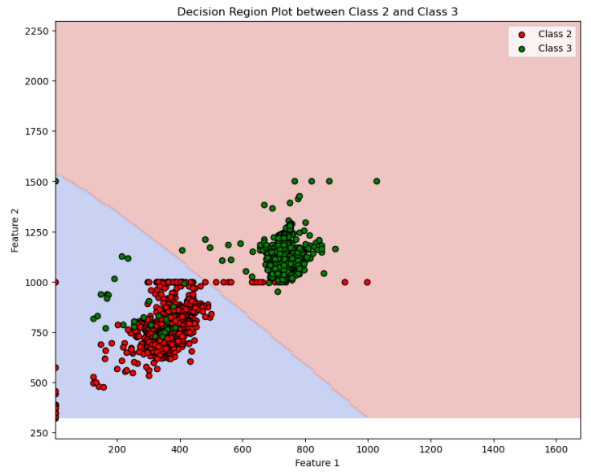


Fig. 18. Decision Region Plot Between Class 2 and Class 3

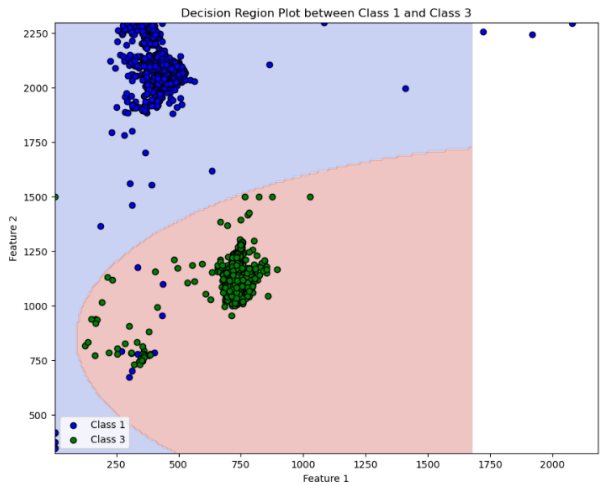


Fig. 19. Decision Region Plot Between Class 1 and Class 3

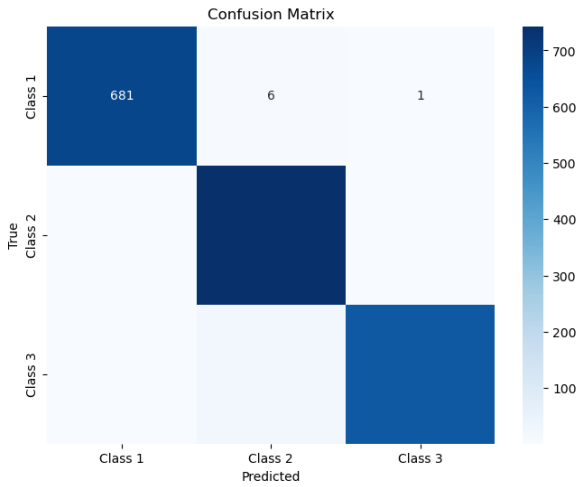


Fig. 20. Confusion Matrix (Full Covariance Matrix for Each Class)

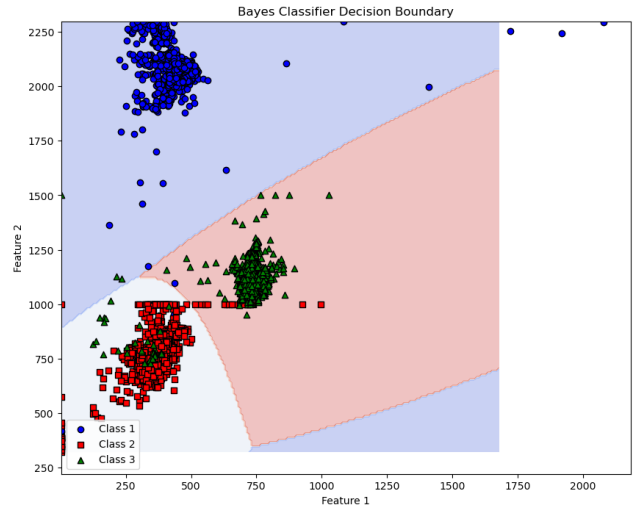


Fig. 22. Decision Region Plot with Training Data

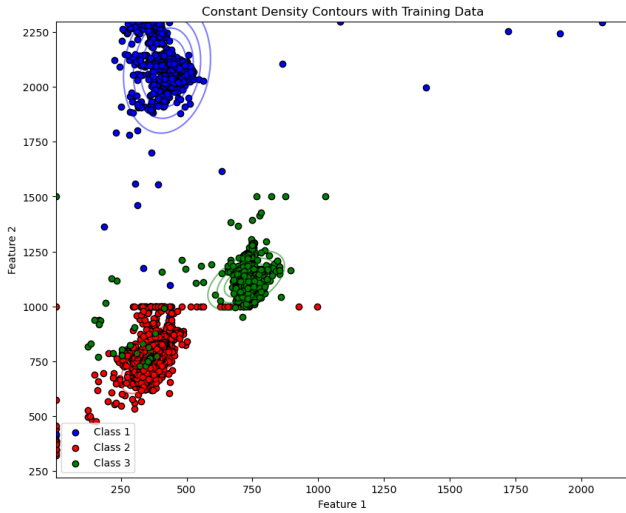


Fig. 21. Constant Density Contours with Training Data

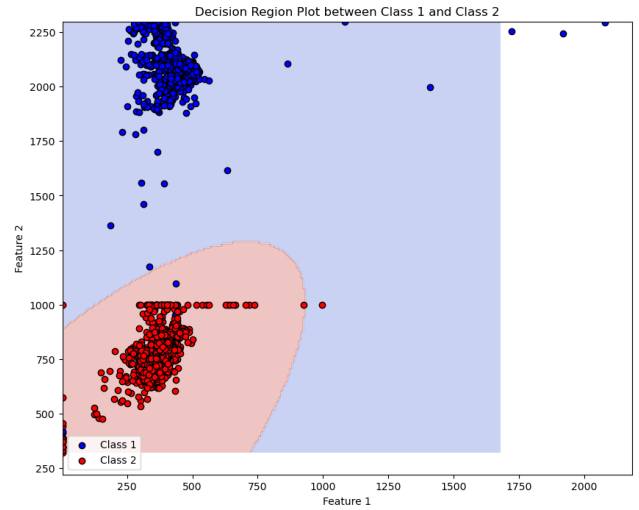


Fig. 23. Decision Region Plot Between Class 1 and Class 2

- **Mean F-Measure:** 0.9842
- **Classifier Accuracy:** 98.42%

XIV. CONCLUSION

In conclusion, the Bayesian classifier was successfully implemented and thoroughly tested on a real-world dataset containing vowel sounds, represented by the formant frequencies F1 and F2, corresponding to the vowel formants /e/, /u/, and /a/. The classifier effectively captured the underlying relationships within the data by leveraging multivariate Gaussian distributions for each class. Through Bayes' theorem, it accurately computed posterior probabilities and assigned each data point to the class with the highest likelihood.

The evaluation metrics, including accuracy, precision, recall, and F1-score, demonstrated the model's strong performance in classifying vowel sounds with minimal errors. The balanced performance across all three classes confirmed the classifier's robustness in handling real-world speech data.

Moreover, the visualizations of decision boundaries provided valuable insights into how the classifier distinguished

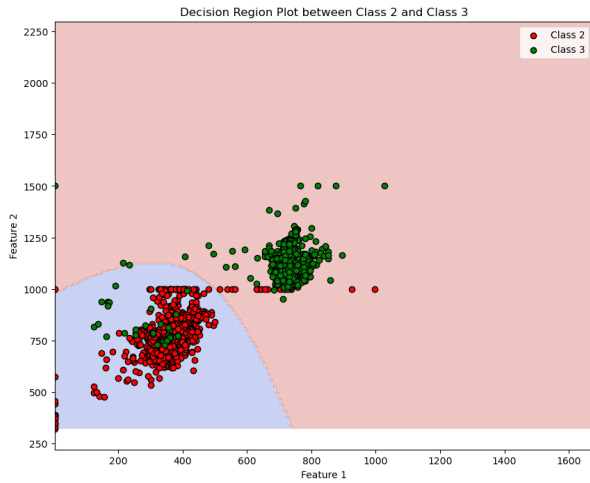


Fig. 24. Decision Region Plot Between Class 2 and Class 3

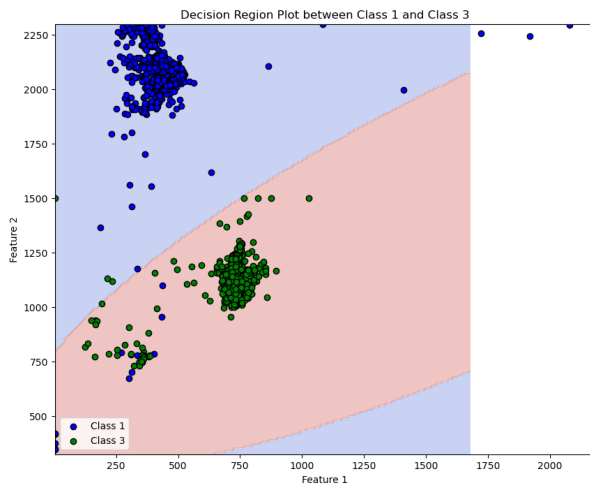


Fig. 25. Decision Region Plot Between Class 1 and Class 3

between the different vowel formants in the feature space. These plots highlighted the dominant regions for each vowel class and illustrated the transitions between classes, offering a clear understanding of the model's decision-making process.

The successful application of the Bayesian classifier to the vowel dataset underscores its potential for real-world speech recognition tasks, where accurate and interpretable classification is essential. The combination of theoretical rigor and visual interpretability demonstrated in this project reaffirms the classifier's effectiveness in statistical pattern recognition.

Future improvements could include:

- Using class-specific covariance matrices to handle cases with differing variances.
- Extending the model to handle higher-dimensional data.

ACKNOWLEDGMENT

The completion of this project would not have been possible without the guidance and support of several individuals. First

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We are also incredibly grateful to all the Teaching Assistants who contributed their time and knowledge. Their assistance with technical concepts, problem-solving, and constructive criticism during the various phases of the project was instrumental in its successful completion. Their willingness to help and share their expertise greatly enriched our learning experience.

Finally, we would like to thank the entire faculty and staff of the Department of Computer Science and Engineering at IIT Dharwad for providing us with the resources and a conducive learning environment that made this project possible.

To everyone who contributed to the success of this project, we extend our heartfelt thanks.

REFERENCES

- [1] Duda, R. O., Hart, P. E., & Stork, D. G. (2000). *Pattern Classification* (2nd ed.). Wiley-Interscience. - A comprehensive textbook covering various classification techniques, including Bayes classifiers and decision theory.
- [2] Bishop, C. M. (2006). *Pattern Recognition and Machine Learning*. Springer. - This book provides a detailed treatment of probabilistic graphical models, including Bayesian classifiers, along with examples and algorithms.
- [3] Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The Elements of Statistical Learning: Data Mining, Inference, and Prediction* (2nd ed.). Springer. - A well-known reference for statistical learning that includes a section on Naive Bayes classifiers and their applications.
- [4] Mitchell, T. M. (1997). *Machine Learning*. McGraw-Hill. - One of the seminal texts in machine learning, providing an introduction to the fundamentals of Bayes classification and probabilistic learning.
- [5] Murphy, K. P. (2012). *Machine Learning: A Probabilistic Perspective*. MIT Press. - A detailed guide to machine learning methods with a probabilistic approach, including extensive discussions on Bayesian classification techniques.